RAED Dynamics · Robot Design Engineering Report

Project **RAED R-1 class · CNC machine tending** · revision **rev-e7daf6eba8** · snapshot b0376b68cbe700c1

1 · Requirements and selected configuration

Item	Value	Item	Value
Application	machine_tending	Robot family	long
Workpiece mass	8.0 kg	Material	al6061
Specified payload	11.0 kg	End effector	rg2 (auto)
Reach requirement	1.300 m	Est. robot mass	54.1 kg
Cycle time target	20.0 s	Path time estimate	2.85 s

2 · Actuator sizing (static worst case)

Axis	Motor / reducer	Worst static (N·m)	Allowed cont. (N·m)	Limited by	Utilization	Required @ 11 kg (N·m)	Static TCP cap. (kg)
J1	sgm7a-08a / csg-32-50 (i=50)	0.0	89.6	motor	0 %	0.0	—
J2	sgm7a-08a / csg-32-50 (i=50)	399.2	89.6	motor	445 %	448.5	0.0
J3	sgm7a-08a / csg-25-80 (i=80)	154.4	75.0	gearbox	206 %	183.3	0.0
J4	sgm7a-04a / csg-25-80 (i=80)	0.0	75.0	gearbox	0 %	0.0	—
J5	sgm7a-04a / csg-17-100 (i=100)	29.3	24.0	gearbox	122 %	39.1	6.4
J6	sgm7a-01a / csg-17-100 (i=100)	0.0	23.9	motor	0 %	0.0	—

Static payload capacity: **0.00 kg** (limiting axis J2). Computed reach: wrist centre **1.355 m**, flange **1.472 m**. Static gravity screening over a J2/J3 sweep with J1, J4 and J6 at zero and J5 swept for the wrist. Continuous ratings only; dynamic torque, acceleration, process force and thermal limits are not included, so real payload capacity is lower.

3 · Benchmark against RAED R-1 target and references

Robot	Role	Payload (kg)	Reach (m)	Mass (kg)	Payload/mass	Evidence
RAED design rev-e7daf6eba8	design	0.00	1.472	54.1	0.000	computed_static_screening
RAED R-1 (IND)	target	11.00	1.300	—	—	public_listing_unverified
Yaskawa Motoman UP6	competitor	6.00	1.373	130.0	0.046	library_header_unverified

Target check	Design	Target	Status	Note
Static payload capacity vs R-1 payload	0.00 kg	11.00 kg	fail	Static upper bound; dynamic capacity is lower.
Computed flange reach vs R-1 reach	1.47 m	1.30 m	pass	R-1 reach definition is not stated by the source.
Static payload vs project requirement	0.00 kg	11.00 kg	fail	
Computed flange reach vs project requirement	1.47 m	1.30 m	pass	

Reference values are public, unverified metadata. Design values are static screening results; neither is a certified rating.

4 · Engineering checks

Check	Status	Method	Reasons
static_torque_J1	pass	recursive gravity with lumped link+actuator masses at one posture	
static_torque_J2	fail	recursive gravity with lumped link+actuator masses at one posture	torque_exceeds_continuous_rating J2
static_torque_J3	fail	recursive gravity with lumped link+actuator masses at one posture	torque_exceeds_continuous_rating J3
static_torque_J4	pass	recursive gravity with lumped link+actuator masses at one posture	

Check	Status	Method	Reasons
static_torque_J5	fail	recursive gravity with lumped link+actuator masses at one posture	torque_exceeds_continuous_rating J5
static_torque_J6	pass	recursive gravity with lumped link+actuator masses at one posture	
dynamic_torque	unknown	not implemented in this release	
deflection	unknown	structural model not in this release	
thermal	unknown	RMS thermal model not in this release	
collision_sampled_aabb	pass	axis-aligned boxes at one or more sampled postures; not a continuous guarantee	
task_Conveyor	pass	DLS IK + FK residual + AABB	
task_Chuck	pass	DLS IK + FK residual + AABB	
task_Out	pass	DLS IK + FK residual + AABB	
payload_definition	pass	explicit payload definition	
payload_requirement	pass	requested load vs specified payload	
family_reach	pass	template nominal reach vs required distance	
payload_static_wrist	fail	mg * TCP lever vs 50% gearbox rated torque at wrist	payload_static_exceeds_wrist
position_accuracy	unknown	needs encoder, compliance, calibration; IK residual is not accuracy	
repeatability	unknown	requires physical test	
cycle_time_path_length	pass	polyline length / TCP speed + 0.4s process per pose; not motor torque/speed curve	

5 · Bill of materials

Part	Description	Make/Buy	Qty	CAD	Unit cost
RB-ASSY-001	Robot assembly long	make	1.0	—	Unknown
RB-BASE-001	Base plate	make	1.0	envelope	Unknown
RB-LNK-01	Link 1	make	1.0	envelope	Unknown
RB-HSG-01	Joint housing 1	make	1.0	envelope	Unknown
RB-LNK-02	Link 2	make	1.0	envelope	Unknown
RB-HSG-02	Joint housing 2	make	1.0	envelope	Unknown
RB-LNK-03	Link 3	make	1.0	envelope	Unknown
RB-HSG-03	Joint housing 3	make	1.0	envelope	Unknown
RB-LNK-04	Link 4	make	1.0	envelope	Unknown
RB-HSG-04	Joint housing 4	make	1.0	envelope	Unknown
RB-LNK-05	Link 5	make	1.0	envelope	Unknown
RB-HSG-05	Joint housing 5	make	1.0	envelope	Unknown
RB-LNK-06	Link 6	make	1.0	envelope	Unknown
RB-HSG-06	Joint housing 6	make	1.0	envelope	Unknown
RB-FLG-001	ISO 9409 tool flange	make	1.0	envelope	Unknown
BUY-SGM7A-08A	0.75 kW AC servo	buy	3.0	missing	Unknown
BUY-CSG-32-50	Strain-wave gearbox ratio 50	buy	2.0	missing	Unknown
BUY-SKF-6206	Deep groove ball bearing 30x62x16	buy	6.0	missing	Unknown
BUY-CSG-25-80	Strain-wave gearbox ratio 80	buy	2.0	missing	Unknown
BUY-SGM7A-04A	0.4 kW AC servo	buy	2.0	missing	Unknown
BUY-CSG-17-100	Strain-wave gearbox ratio 100	buy	2.0	missing	Unknown
BUY-SGM7A-01A	0.1 kW AC servo	buy	1.0	missing	Unknown
BUY-ISO4762-M6X20	ISO 4762 M6x20	buy	24.0	missing	Unknown
EE-RG2	Parallel gripper RG2	buy	1.0	envelope	Unknown

Part	Description	Make/Buy	Qty	CAD	Unit cost
CON-LOCTITE-243	Threadlocker (consumable)	buy	1.0	not_applicable	Unknown

6 · Limitations

Static gravity screening only: dynamic torque, deflection, thermal behaviour, cable routing and physical accuracy remain unverified. Reference robot values are public, unverified metadata. Unknown values are reported as unknown rather than estimated.